

Dr Benya Wang

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CAREER PROFILE

Senior AI and data scientist with a PhD and 8 years delivering production AI across mining and resources — end to end, from data exploration and feature engineering through model development, deployment, and optimisation. Works daily with complex operational data (time-series, sensor, and process historian) and has taken Generative AI to production at enterprise scale: RAG knowledge bases, agentic multi-agent workflows, and LLM applications on Azure (AI Foundry, Azure OpenAI, AI Search, Azure ML) with Databricks and MLflow. Builds repeatable, well-documented AI workflows, applies responsible AI through governance frameworks designed with Cyber and Enterprise Architecture, and delivers alongside operational teams across Australia, Africa, and South America.

CORE COMPETENCIES

Machine Learning & Statistical Modelling

Predictive analytics, time-series and sensor data modelling, feature engineering, XGBoost, LightGBM, Random Forest, computer vision (YOLO), advanced process control

Generative AI & Agentic Systems

RAG pipeline design, agentic and multi-agent workflows, Azure OpenAI, Azure AI Foundry, Azure AI Search, LangChain, LightRAG, GraphRAG, AutoGen, vector stores, Microsoft Copilot Studio, Claude Code

Azure & Data Platforms

Azure (AI Foundry, Azure ML, AI Search, IoT Edge/Hub, ACR), Databricks (Feature Store, workflow orchestration, data warehouse), AWS

MLOps & Responsible AI

MLflow, model endpoint deployment, AKS / ACI, Docker, Azure DevOps / Pipelines, CI/CD pipeline design, repeatable and documented AI workflows, AI governance frameworks

Engineering & Delivery

Python, PySpark, Pandas, SQL, stakeholder engagement, technical mentoring, cross-site delivery across multiple time zones

PROFESSIONAL EXPERIENCE

Specialist Machine Learning Engineer

Apr 2023 – Present

South32 · Perth, WA

- Built a safety chatbot and GenAI knowledge base over safety and geoscience corpora (Azure AI Foundry, Azure AI Search, LightRAG) end to end — corpus exploration, chunking strategy, retrieval tuning, and production deployment — giving operational teams conversational access to guidance locked in unstructured documents.
- Defined the technical strategy and built a unified AI platform (Azure AI Search, Azure OpenAI, Cosmos DB) with self-service patterns and reference implementations that reduced time-to-production for new LLM applications by ~60%, enabling multiple product teams to ship independently.
- Pioneered an air-gapped edge ML deployment pattern (Azure IoT Edge, Databricks Apps) enabling model inference on isolated industrial sites with zero internet connectivity; adopted as the standard across all remote operations.
- Owned model deployment and optimisation, right-sizing the container runtime for inference endpoints — ACI for lightweight, single-service workloads and AKS where orchestration and scaling were required — packaging models as Docker microservices published through ACR.
- Applied responsible AI in practice: partnered with Cyber and Enterprise Architecture to design a Microsoft Copilot Studio governance framework and a GitHub / GitHub Copilot governance framework, establishing enterprise-wide standards for AI-assisted development, access controls, and compliant deployment.
- Authored the standards, engineering guides, and platform patterns that made AI delivery repeatable and well documented across the business, and mentored data scientists across business units to drive consistent, production-quality outcomes.

- Served as Acting Data Science Lead May–Jun 2026, maintaining delivery continuity across active AI workstreams; co-designed the production support model with the Operational Digital Technology team, defining ownership, escalation paths, and SLA structures.

Specialist Data Scientist

Jan 2022 – Apr 2023

South32 · Perth, WA

- Delivered advanced process control solutions over plant time-series and sensor data for operations in Australia, Africa, and South America, working with local technology teams in each region to improve recovery and energy efficiency — coordinating delivery across multiple time zones.
- Built reusable PySpark feature pipelines into a Databricks Feature Store over large-scale industrial sensor and historian data, replacing ad-hoc notebook preparation with versioned, model-ready features.
- Implemented production multi-agent LLM pipelines (GraphRAG, AutoGen) to extract structured insights from HR records, safety incident reports, and geoscience exploration corpora.
- Architected and led delivery of a centralised Data Science platform (Azure Databricks, MLflow, Azure Pipelines) that standardised MLOps workflows across 6+ business units, cutting the average model deployment cycle from ~3 weeks to under 5 days and eliminating ad-hoc deployment risk.
- Designed and deployed a Vision AI safety system (YOLO, Supervision) for automated stop-sign monitoring, directly improving operational safety compliance.

Data Scientist

Jan 2021 – Jan 2022

Rio Tinto · Perth, WA

- Engineered serverless data pipelines (AWS Lambda, Glue, Athena) for real-time production analytics supporting operational decision-making.
- Led ML-driven optimisation of ore blending strategies, improving resource utilisation and reducing annual operational costs.
- Won an internal Hackathon for diamond image classification using TensorFlow and transfer learning.

Graduate Data Scientist

Feb 2019 – Jan 2021

Rio Tinto · Perth, WA

- Developed predictive maintenance models (LightGBM, Random Forest) for a 1,700 km rail network, reducing unplanned downtime through early fault detection.
- Containerised models in Docker and deployed via AWS SageMaker, establishing repeatable deployment and rollback processes for the team.

EDUCATION

PhD in Environmental Engineering

Feb 2015 – Feb 2019

University of Western Australia

Thesis: Application of machine learning algorithms in urban catchments. Five peer-reviewed publications in *Geoscientific Model Development*, *Water Resources Research*, and other journals.

Bachelor of Science

Sep 2010 – Jun 2014

Zhejiang University, China

CERTIFICATIONS

Databricks Certified

Machine Learning Professional · Machine Learning Associate · Data Engineer Associate

Microsoft Certified

DevOps Engineer Expert · Azure Data Scientist Associate · Azure Data Engineer Associate · Azure Developer Associate

AWS Certified

Machine Learning – Specialty · Solutions Architect – Associate

Other

HashiCorp Certified: Terraform Associate · Anthropic: Claude Code in Action